Injury 56 (2025) 112660

Contents lists available at ScienceDirect

Injury

journal homepage: www.elsevier.com/locate/injury

Effect of electroacupuncture intervention before and after operation on perioperative neurocognitive disorders in elderly patients with hip fractures: A randomized controlled trial

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ARTICLE INFO

Keywords:
Electroacupuncture
Transcutaneous electrical acupoint stimulation
Hip surgery
Perioperative neurocognitive disorders
Pain
Postoperative nausea and vomiting

ABSTRACT

Introduction: The incidence of postoperative neurocognitive disorder (PND) in elderly patients with hip fractures poses a significant clinical challenge, with current management strategies offering limited efficacy in prevention or resolution. This prospective study evaluated the effectiveness of pre-and postoperative electroacupuncture (EA) intervention in mitigating PND in this patient cohort.
Methods: A double-masked, randomized controlled trial was conducted involving 60 elderly patients (≥65 years) with fragility hip fractures scheduled for surgical repair. Participants were randomly assigned to either the EA intervention group (Group A) or a non-stimulated control group (Group C). Mini-Mental State Examination (MMSE) scores were recorded at baseline and 1, 3, and 7 days postoperatively, while ELISA was used to assess IL-1β, IL-6, and S-100β levels. Time-varying MAP, SpO2, and HR were measured. Adverse cardiovascular events, extubation duration, recovery room stay, VAS scores, analgesia pump use, postoperative adverse responses, and hospitalization length were recorded.
Results: Among 60 randomized patients (mean age 74.02 years; 54.7 % male), 53 were analyzed for primary outcomes. Postoperative day 1 PND incidence was significantly lower in Group A (25.0 %) than Group C (56.0 %; $P < 0.05$), persisting on day 3 (Group A: 14.3 %, Group C: 48.0 %; $P < 0.05$). By day 7, PND incidence was similar in both groups. Time-group interactions were significant for IL-1β, IL-6, and blood pressure ($P < 0.05$). Group A exhibited a lower VAS score at 24 h postoperatively (2.65 ± 0.94 vs. 3.96 ± 0.96; $P < 0.05$). Adverse events were reported in 26 Group A and 32 Group C cases. Postoperative nausea and vomiting (PONV) significantly differed (Group A: 3.7 %, Group C: 30.8 %).
Conclusions: The findings suggest that pre- and postoperative EA stimulation may significantly reduce the risk of PND, modulate inflammatory responses, and lower blood pressure. Furthermore, EA intervention was associated with reduced postoperative pain and a marked decrease in the incidence of PONV in elderly patients with hip fractures. These results highlight the potential therapeutic benefits of EA in managing PND in this vulnerable patient population and warrant further investigation.
Subject words: electroacupuncture, transcutaneous electrical acupoint stimulation, hip surgery, perioperative neurocognitive disorders, pain, postoperative nausea and vomiting

Introduction

Hip fractures in the geriatric population represent a widespread and concerning health issue, commanding considerable international attention [1]. These fractures can lead to profound adverse outcomes for affected individuals. Surgery, a primary therapeutic modality, frequently triggers systemic inflammatory and stress responses, which may intensify neuronal inflammation and diminish cognitive function in

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https://doi.org/10.1016/j.injury.2025.112660
Accepted 2 August 2025
Available online 7 August 2025

elderly patients [2]. Postoperative neurocognitive disorder (PND) is characterized by a decline in cognitive domains such as memory, attention, executive function, and information processing speed in the perioperative period compared to preoperative levels [3]. The prevalence of PND has been documented to be between 25.8 % and 41.4 % at one week and between 9.9 % and 12.7 % at three months post-noncardiac surgery in individuals over the age of 65 [4]. This cognitive deterioration can persist for extended periods, potentially leading to increased caregiver dependence, reduced quality of life, an elevated risk of dementia, and negative impacts on long-term survival. Although the etiology of PND remains to be fully elucidated, the neuroinflammatory response is widely recognized as a critical factor in its pathogenesis [5].

Transcutaneous electrical acupoint stimulation (TEAS) is a comprehensive therapy that combines the conduction of meridians and acupoints with electrical stimulation, belonging to the electroacupuncture (EA) category. It can generate various parameters such as waveform, duration, frequency, and intensity to achieve different effects. Studies have demonstrated that TEAS can alleviate perioperative tension in surgical patients, enhance cerebral blood circulation, and mitigate brain tissue damage to benefit PND [6–8]. Moreover, it exerts a specific inhibitory effect on the NF-kB signal transduction pathway, reducing Interleukin-6 (IL-6) and tumor necrosis factor-alpha (TNF-α) levels in central nervous cells while promoting functional recovery [9]. This study aims to assess the impact of EA intervention at Baihui (GV20), Neiguan (PC6), Zusanli (ST36), and Sanyinjiao (SP6) acupoints on PND occurrence in elderly hip fracture patients. By applying EA stimulation pre- and postoperatively, we focus on its effects on perioperative hemodynamic stability, adverse reactions, postoperative pain, and hospital stay duration. This research is significant as it may provide a novel, non-pharmacological approach to facilitate the swift recovery of PND in geriatric patients and potentially reduce healthcare expenses.

Material and methods

We conducted this randomized, prospective, double-blind, intervention-controlled clinical trial in the Second Affiliated Hospital of Zhejiang Chinese Medical University Xinhua Hospital. The study was approved by the appropriate Institutional Review Board (IRB), and written informed consent was obtained from all participants. This study was approved by the Ethics Committee of the Second Affiliated Hospital of Zhejiang Chinese Medicine University (IRB: 2019-KL-129–01). The trial was registered before patient enrollment at clinicaltrials.gov (see https://www.chictr.org.cn/showproj.html?proj=136703, principal investigator: Wei Chen, date of registration: November 10, 2021) in the China Clinical Trial Registration Center (registration number: ChiCTR2100053050).

Participants

All patients provided written informed consent prior to enrollment in this study. The inclusion criteria were as follows: (1) patients with hip fracture included femoral neck fracture, femoral intertrochanteric fracture, and acetabular fracture; (2) age≥65 years old; ASA classification I~III; (3) Mini-Mental State Examination (MMSE) scores ≥ 24 for middle school graduates, ≥ 20 for primary school graduates, and ≥ 17 for illiterate individuals. The exclusion criteria included: (1) severe visual or hearing impairment, or an inability to communicate effectively; (2) having senile dementia; (3) having a history of epilepsy, alcoholism, neurological diseases, and mental illness; (4) having an infected acupoint on the skin.

Sample size calculation

The sample size for this study was calculated using PASS15.0 software. A previous study reported that the incidence of PND ranges from 10 % to 60 % in orthopedic surgery [10]. Based on our initial experiment, the occurrence of PND decreased from 45 % to 20 % following TEAS treatment. The statistical power was 80 %, and the significance threshold of 0.05 for a two-sided test. The sample size was determined to be 50. Considering the dropouts and refusals to participate in the follow-up, the sample size was increased by 20 %. Consequently, 60 patients who underwent elective hip fracture surgery at the Second Affiliated Hospital of Zhejiang Chinese Medicine University between December 2021 and August 2022 were included in the study.

Research grouping

Patients who met the inclusion criteria were randomly assigned to either the control group (Group C) or the electroacupuncture group (Group A), with 30 participants in each group using a random number table method. In Group C, electrodes were placed at the acupuncture points without electrical stimulation. In contrast, patients in Group A received electroacupuncture stimulation, administered by certified acupuncturists. The stimulation occurred 30 min before anesthesia on the day of surgery and 30 min post-anesthesia in the post-anesthesia care unit (PACU) when the recovery score was ≥9.

Anesthesia management

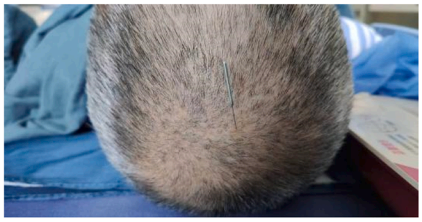
All patients received general endotracheal anesthesia (GETA) with total intravenous anesthesia (TIVA). Anesthesia induction was performed using etomidate 0.3mg/kg, sufentanil 0.3–0.5 μg/kg, and cisatracurium 0.15–0.2 mg/kg. Following endotracheal intubation, mechanical ventilation was initiated. The anesthetic was maintained with intravenous propofol (4–10 mg/kg/h), remifentanil (0.2–2 ug/kg/min), and cisatracurium (intermittently) to relax muscles. During the surgery, the bispectral index was maintained between 40 and 60, while heart rate and blood pressure were kept within 20 % of baseline values.

Postoperative analgesia

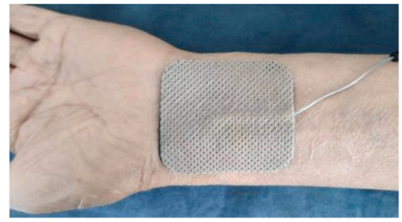
All patients received postoperative patient-controlled intravenous analgesia (PCIA). The PCIA regimen consisted of 2 μg/kg sufentanil diluted in 100 ml of saline. The parameters were set: flow rate, 1.5 ml/h; bolus, 2 ml; lockout time, 15 min.

Intervention

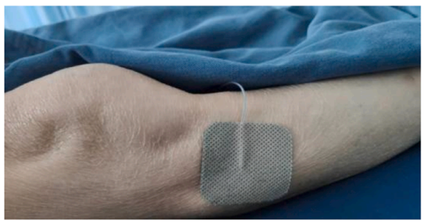
Acupoints were located according to the People's Republic of China standard "Acupoint Name and Positioning" (GB/T12346–2006). Baihui (GV20): at the intersection of the line connecting the apexes of the two auricles and the median line of the head, 7 cun directly above the posterior hairline and 5 cun behind the anterior hairline; Neiguan (PC6): on the anterior forearm, two transverse finger widths above the wrist crease, between the tendons of palmaris longus and flexor carpi radialis; Zusanli (ST36): on the anterior aspect of the lower leg, 3 transverse finger widths below the knee, one finger-breadth from the anterior crest of the tibia; Sanyinjiao (SP6): on the inner side of the lower leg, 3 transverse finger widths above the inner ankle, on the posterior border of the tibia. Patients in Group A received 30-minute electrical stimulation at Baihui (GV20), bilateral Neiguan (PC6), Zusanli (ST36), and Sanyinjiao (SP6) before and after anesthesia (PACU awakening score ≥ 9). See Fig.#x00A0;1. The Huatuo brand electronic pulse therapeutic device (SDZ-II, Suzhou Medical Products Factory Co., LTD., Suzhou, China) was used to stimulate acupoints with a millimeter needle or electrode patch. The stimulation parameters were set as follows: 2/100 Hz density wave, adjusted to patient tolerance, with each session lasting 30 min.



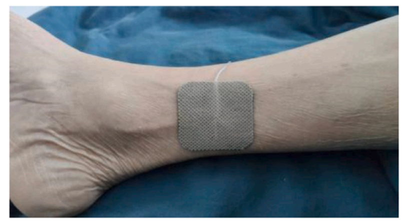
Baihui (GV20): at the intersection of the line connecting the apexes of the two auricles and the median line of the head, 7 cun directly above the posterior hairline and 5 cun behind the anterior hairline.



Neiguan (PC6): on the anterior forearm, two transverse finger widths above the wrist crease, between the tendons of palmaris.



Zusanli (ST36): on the anterior aspect of the lower leg, 3 transverse finger widths below the knee, one finger-breadth from the anterior crest of the tibia.



Sanyinjiao (SP6): on the inner side of the lower leg, 3 transverse finger widths above the inner ankle, on the posterior border of the tibia.

Fig. 1. The location of the acupoints for transcutaneous electrical acupoint stimulation.

Indicators of observation

General feature comparison

The general characteristics included age, sex, body mass index (BMI), level of education, American Society of Anesthesiologists (ASA) grade, number of previous surgeries, duration of anesthesia, surgical procedure time, volume of fluid rehydration, amount of blood loss and transfusion required, number of drainage tubes used, lost hemoglobin, extubation time after surgery completion, duration in the recovery room following surgery completion, length of hospital stay, combined underlying diseases and high-risk factors for both groups of patients.

Monitoring vital signs at different time points

Patients' mean arterial pressure (MAP), heart rate (HR), and oxygen saturation (SpO2) were monitored at various time points, including upon admission (T0), before anesthesia induction (T1), during intubation (T2), before skin incision (T3), skin incision (T4), 30 min after skin incision (T5), 60 min after skin incision (T6), completion of surgery (T7), and upon leaving the recovery room (T8).

Monitoring of hemodynamic stability criteria

Hypotension (MAP $<$ 80 %) or hypertension (MAP $>$ 120 %) was recorded if the intraoperative MAP was below or above 20 % of the admission MAP for 5 min or more; when the HR is $<$48 bpm or $>$90 bpm and lasts for 5 min, recorded bradycardia (HR$<$48 bpm) and tachycardia (HR$>$90 bpm).

Neuropsychiatric function tests

The Mini-Mental State Examination (MMSE) was used to assess cognitive function. It was performed one day before and 1, 3, and 7 days after surgery. When the MMES score dropped two or more points after surgery, PND was considered [11].

PCIA using and pain score

The Visual Analog Scale (VAS) pain score was documented at 24 and 48 h post-surgery. The VAS score ranged from 0 to 10, indicating that the pain was more severe as the score increased. The dosage of the PCIA medication and pressing times were recorded 48 h post-surgery.

Observation of postoperative adverse reactions

Patients' surgical complications were recorded. Postoperative nausea and vomiting (PONV), abdominal discomfort, diarrhea, and other digestive disturbances were noted. The cardiovascular system included postoperative hypertension, hypotension, arrhythmia, bradycardia, tachycardia. Respiratory symptoms included hypoxemia (SpO2$<$90 %) and pulmonary infection. Central nervous system symptoms included headache, dizziness, confusion, restlessness, chills, and fever. The adverse reactions of the two groups were compared.

Detecting of serum indicators

Peripheral blood samples (3 ml) were collected 30 min before anesthesia (t0), 30 min after surgery (t1), and one day after surgery (t2). The supernatant was collected and refrigerated at −80 °C after 10 min centrifugation at 1300 RPM. Serum levels of IL-1β, IL-6, and S100-β protein were measured using an ELISA kit (Jiangsu Enzyme Immunoassay Industry Co. Ltd, Yancheng, China).

Statistical analyses

Data were analyzed using SPSS 25.0 statistical software (Version 25, IBM Corporation, Armonk, NY, USA). The mean±SD was employed to represent measurement data that followed a normal distribution, while categorical variables were presented as cases and percentages. Intergroup comparisons were performed using the Kruskal-Wallis test; average distribution data were compared using the Independent-Sample T-Test, and repeated-measurement data were analyzed through repeated-measures ANOVA. Counting data was assessed using the X^2-test. $P < 0.05$ was considered statistically significant.

Results

Clinical characteristics of patients and surgical results

A total of 60 patients who met the inclusion and exclusion criteria were initially enrolled. Two patients withdrew consent, one canceled surgery, and four were lost to follow-up. Ultimately, 53 patients were included in the trial. The patients were randomly assigned to Group C (n = 26) and Group A (n = 27) using a random number table (Fig.

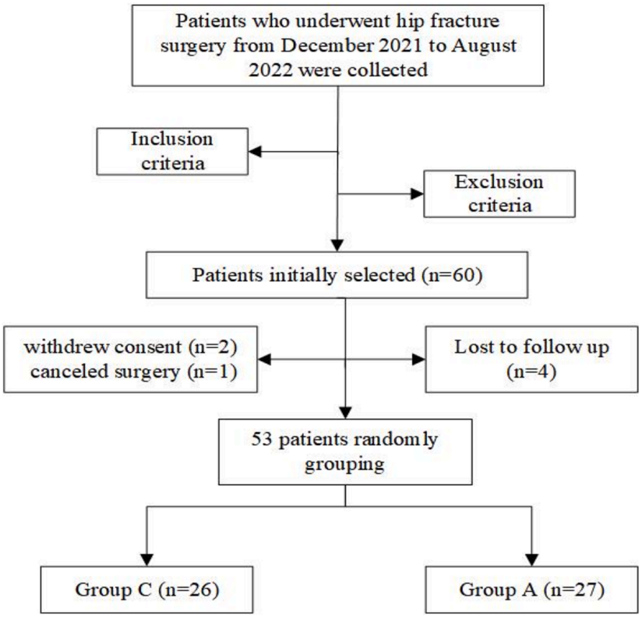


Fig. 2. Experimental flow of this study.

#x00A0;2). The two groups did not differ in age, sex, number of previous operations, BMI index, ASA categorization, education level, or combined underlying disorders ($P > 0.05$). Anesthesia duration, operation time, blood loss, blood transfusion, infusion volume, number of drainage tubes implanted after surgery, and hemoglobin loss were not significantly different across groups ($P > 0.05$), Table 1. Thus, both groups were comparable.

Table 1
Comparison of general conditions of patients in each group ($n = 53$).

Characteristics	C ($n = 26$)	A ($n = 27$)	T-test/ X^2-test	
			t/X^2	P
Age [years, mean (SD)]	72.35 ±15.15	75.70±10.95	−0.927	0.358
sex [n (%)]				
Female	11 (42.3 %)	13 (48.1 %)	0.182	0.669
Male	15 (57.7 %)	14 (51.9 %)		
History of operations [times, mean (SD)]	0.65±0.69	0.81±0.79	−0.791	0.432
BMI [kg/m^2, mean (SD)]	22.22±2.53	21.61±2.58	0.876	0.385
ASA [n (%)]				
I	1 (3.8 %)	1 (3.7 %)	5.794	0.98
II	17 (65.4 %)	17 (63.0 %)		
III	8 (30.8 %)	9 (33.3 %)		
Educational level [n (%)]				
Primary school	11 (42.3 %)	10 (37.0 %)	5.664	0.129
Middle school	8 (30.8 %)	3 (11.1 %)		
High school	2 (7.7 %)	7 (25.9 %)		
College	5 (19.2 %)	7 (25.9 %)		
Combined underlying diseases [n (%)]				
Hypertension	12 (46.2 %)	15 (55.6 %)	0.468	0.494
Diabetes	6 (23.1 %)	4 (14.8 %)	0.591	0.442
Cardiopathy	3 (11.5 %)	6 (22.2 %)	0.448	0.503
Cerebrovascular	2 (7.7 %)	2 (7.4 %)	0.002	0.969
Anesthesia time [minutes, mean (SD)]	229.15 ±114.15	204.44 ±145.99	0.685	0.497
Surgical duration [minutes, mean (SD)]	196.85 ±113.16	166.19 ±135.86	0.891	0.377
Blood loss [ml, mean (SD)]	183.85 ±122.25	151.11 ±128.13	0.951	0.346
Number of blood transfusions [n (%)]	11 (42.3 %)	10 (37.0 %)	0.154	0.695
Infusion volume [ml, mean (SD)]	2122.31 ±761.94	1764.81 ±1084.52	1.384	0.172
Number of drainage tubes [mean (SD)]	1.08±1.23	0.85±1.20	0.674	0.503
Lost hemoglobin [g/dl, mean (SD)]	1.36±1.04	1.31±0.90	0.175	0.862

Primary outcome

Comparison of MMSE scores and PND incidence

Before surgery, the two groups had similar MMSE scores. The MMSE scores of both groups reduced significantly after surgery (1 and 3 days) ($P < 0.05$), but there were no significant differences between the two groups on postoperative day 7 ($P > 0.05$). Both groups experienced a significant increase in MMSE scores on postoperative day 7 compared to postoperative day 1 ($P < 0.05$). The incidence of PND on postoperative day 1 was 56.0 % in group C and 25.0 % in group A ($P < 0.05$). On postoperative day 3, the incidence of PND was 48.0 % in Group C and 14.3 % in Group A ($P < 0.05$). The incidence of PND was similar between the two groups on postoperative day 7 ($P > 0.05$). See Table 2 and Fig. #x00A0;3, 4.

Comparison of serum indicators

Repeated measures ANOVA for IL-1β revealed that time-group interaction was significant ($F = 7.827$, $P = 0.002$). IL-1β levels significantly increased in group C at t1 and t2 ($P < 0.05$) and in group A at t1 ($P < 0.05$) compared to t0. There was no significant difference between t2 and t0 ($P > 0.05$). Group A did not show a significant increase in IL-1β levels at t2 compared to t1 ($P>0.05$).

Repeated measures ANOVA for IL-6 also showed that interaction between time and group was significant ($F = 9.460$, $P = 0.001$). At t1, IL-6 levels were significantly lower in Group A compared to Group C ($P<0.05$). IL-6 levels in Group A were similar at t1 and t2 compared to t0 ($P > 0.05$). Group C exhibited a significant decrease in IL-6 levels at t2 compared to t1 ($P<0.05$).

Repeated measures ANOVA for S-100β indicated that time-group interaction was insignificant ($F = 0.545$, $P = 0.586$). At t1, S-100β levels considerably rose in both groups compared to t0 ($P<0.05$). At t2, S-100β levels were significantly lower in both groups compared to t1 ($P < 0.05$). No significant differences in S-100β levels were observed between the groups at any time point($P < 0.05$). See Table 3 and Fig. #x00A0;5.

Secondary outcomes

Comparison of intraoperative hemodynamic stability criteria

Repeated measures ANOVA for MAP revealed that time-group interaction was significant ($F = 3.355$, $P = 0.004$). At T2, Group A exhibited a significantly lower MAP value compared to Group C ($P < 0.05$). Both groups experienced significant MAP declines at T3, T4, T5, and T6 compared to baseline levels at T0 ($P < 0.05$).

Repeated measures ANOVA for HR showed that time-group

Table 2
Comparison of MMSE score and PND incidence between two groups at different times ($n = 53$, $\bar{x}\pm s$).

Group	pre-operation	1st day after surgery	3rd day after surgery	7th day after surgery
MMSE scores				
C	25.30±3.80	23.77±3.98[a]	24.00±3.82[a]	25.00±3.89[b,c]
A	25.74±4.37	24.89±4.68[a]	25.00±4.58[a]	25.74±4.60[b,c]
The incidence of PND (%)				
C	/	14 (56.0 %)	12 (48.0 %)	2 (8.0 %)
A	/	7 (25.0 %) [d]	4 (14.3 %) [d]	1 (3.6 %)

Note:.
[a] , compared with preoperative, $P < 0.05$;.
[b] , compared with 1st day after surgery, $P < 0.05$;.
[c] , compared with 3rd day after surgery;.
[d] , compared with group C, $P < 0.05$.

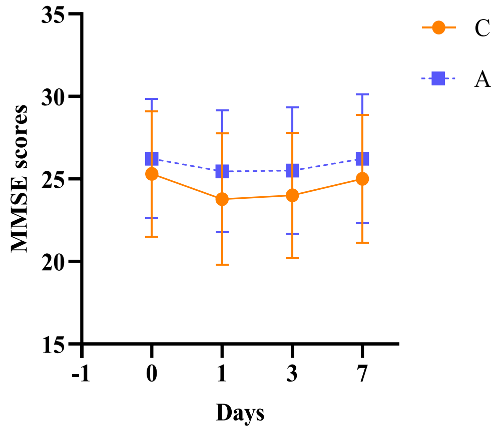


Fig. 3. Comparison of MMSE scores before surgery and 1, 3, and 7 days after surgery between the two groups.

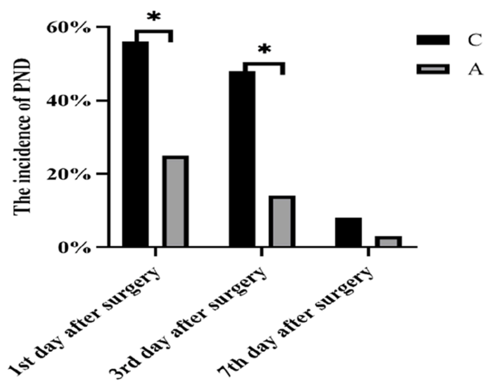


Fig. 4. Comparison of PND incidence between two groups at different times. Note: Compared with group C, * $P < 0.05$.

interaction was insignificant ($F = 1.238$, $P = 0.301$). Both Group C and Group A exhibited significant decreases in HR at T4, T5, and T6 compared to baseline ($P < 0.05$). Additionally, repeated measures ANOVA for oxygen saturation (SpO2) indicated no significant time-group interaction ($F = 0.520$, $P = 0.835$), See Table 4 and Fig.#x00A0;6.

Comparison of intraoperative hemodynamic stability and postoperative recovery between the two groups

Intraoperative hypotension occurred in 50.0 % of patients in Group C and 33.3 % in Group A ($P = 0.735$). Intraoperative hypertension was observed in 15.4 % of patients in Group C and 22.2 % in Group A, with no significant difference between the groups ($P = 0.329$). Intraoperative bradycardia was noted in 11.5 % of patients in Group C and 11.1 % in Group A ($P = 0.375$). The incidence of intraoperative tachycardia was 7.7 % in Group C and 22.2 % in Group A ($P = 0.083$). There was no significant difference in the mean extubation time between Group C (10.85±8.44 min) and Group A (7.44±6.03 min) ($P = 0.097$). Additionally, the proportion of patients with postoperative recovery room stays exceeding 30 min was higher in Group C (42.3 %) than in Group A (29.6 %), although this difference was not statistically significant ($P = 0.336$). See Table 5 and Fig.#x00A0;7.

Comparison of postoperative adverse reactions between the two groups

The incidence of postoperative nausea and vomiting (PONV) was significantly lower in Group A (3.7 %) compared to Group C (30.8 %) (X^2=5.097, $P = 0.024$). There were no differences between the two groups in terms of stomach pain and distension, postoperative hypotension, tachycardia, bradycardia, hypoxemia, headache and vertigo, delirious agitation, chills, and fever (P>0.05). See Table 6 and Fig. #x00A0;8.

Comparison of pain levels and duration of hospitalization

The Visual Analog Scale (VAS) pain score at 24 h post-operation was significantly higher in Group C (3.96±0.96) compared to Group A (2.65 ±0.94) ($t = 4.458$, $P < 0.001$). At 48 h post-operation, the VAS score remained significantly higher in Group C (2.65±0.94) compared to Group A (1.85±0.72) ($t = 2.876$, $P = 0.001$). The mean number of patient-controlled intravenous analgesia (PCIA) compressions at 48 h post-surgery was significantly higher in Group C (8.62±1.84) compared to Group A (7.11±2.14). At 48 h, the two groups had significantly different mean PCIA compressions ($t = 2.745$, $P = 0.008$). Similarly, the mean consumption of PCIA at 48 h post-surgery was significantly higher in Group C (89.23±3.67 ml) compared to Group A (86.22±4.27 ml) ($t = 2.745$, $P = 0.008$). The average hospital stay was 14.77±3.57 days for Group C and 13.37±3.03 days for Group A, with no significant difference the groups ($t = 1.541$, $P = 0.13$). See Table 7 and Fig.#x00A0;9.

Analysis and discussion

Hip fracture surgery is highly invasive, and despite advancements in rapid postoperative recovery strategies, the risk of complications remains high, with central nervous system complications such as PND being prevalent. The underlying mechanisms of PND have yet to be fully elucidated and may involve central inflammation [12,13], decreased cholinergic system function [14], synaptic dysfunction [15], protein dysfunction [16], and gut microbiota imbalance [17]. Among these, the mechanism of central inflammatory response is particularly critical and

Table 3
Comparison of serum levels of IL-1β, IL-6 and S-100β at different time points between the two groups ($n = 53$, $\bar{x}\pm s$).

Group	t0	t1	t2	Repeated-measures ANOVA					
				Group main effect		Time main effect		Group*Time	
				F	P	F	P	F	P
IL-1β (pg/ml)									
C	30.45±5.27	38.27±5.80[b]	36.18±6.49[b]	0.386	0.540	30.851	<0.001	7.827	0.002
A	33.34±6.44	35.74±6.28[b]	32.11±6.09[c]						
IL-6 (pg/ml)									
C	40.42±9.68	53.26±8.25[b]	39.50±11.38[c]	3.974	0.056	22.684	<0.001	9.460	0.001
A	37.22±7.80	40.18±9.85[a]	37.36±9.47						
S-100β (ng/ml)									
C	0.56±0.05	0.82±0.08[b]	0.63±0.18[c]	0.549	0.447	107.034	<0.001	0.545	0.586
A	0.56±0.09	0.77±0.09[b]	0.60±0.20[c]						

Note:.
[a] , compared with group C, $P < 0.05$;.
[b] , compared with t0, $P < 0.05$;.
[c] , compared with t1, $P < 0.05$. t0: before induction of anesthesia (before EA), t1:30 min after surgery (after EA), t2: 1 day after surgery.

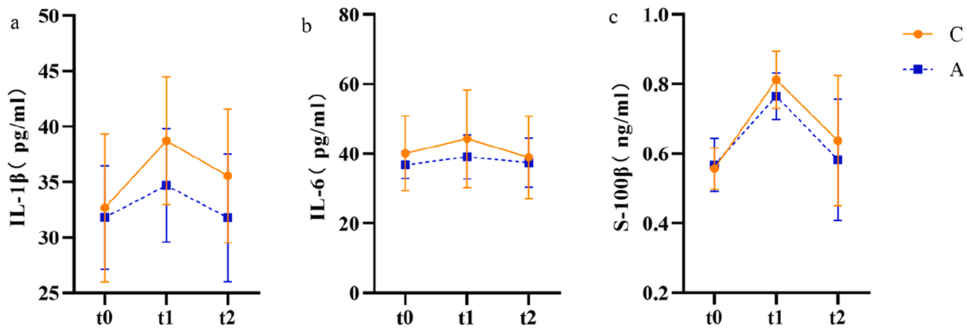


Fig. 5. Comparison of serum levels of IL-1β, IL-6, and S-100β at t0, t1, and t2 between the two groups
Note: Panel a shows the changes in serum IL-1β levels at different times, panel b shows the changes in serum IL-6 levels at different times, and panel c shows the changes in serum S-100β levels at different times.

Table 4
Comparison of MAP, HR, and SpO2 between the two groups at different time points during operation ($n = 53, \bar{x} \pm s$).

Time	C	A	Repeated-measures ANOVA					
			Group main effect		Time main effect		Group*Time	
			F	*P*	*F*	*P*	*F*	*P*
MAP (mmHg)								
T0	102.65±14.02	98.22±11.13	2.539	0.117	14.940	<0.001	3.355	0.004
T1	99.73±15.31	92.37±13.15						
T2	97.85±19.87	80.11±15.87[a,b,h]						
T3	81.27±19.41[a,b,c]	77.44±16.65[a,b]						
T4	90.27±17.61[a]	83.15±17.35[a]						
T5	82.50±15.21[a,b,c]	76.41±14.52[a]						
T6	83.85±13.10[a,b,c]	81.78±13.64[a,b]						
T7	93.62±16.46	99.19±16.47[c,d,e,f,g]						
T8	95.23±14.28 g	99.81±13.01[c,d,e,f,g]						
HR (bpm)								
T0	76.19±16.36	76.37±10.01	0.891	0.350	9.837	<0.001	1.238	0.301
T1	73.37±16.28	77.11±13.73						
T2	74.15±20.03	76.96±14.93						
T3	68.50±19.53	71.04±15.98						
T4	65.85±16.11[a]	71.04±15.30						
T5	63.65±15.10[a,b]	66.19±13.02[a,b,e]						
T6	61.81±17.24[a,b,c]	65.85±13.75[a,b]						
T7	66.77±15.11	74.04±17.00						
T8	69.46±13.82	69.15±17.54						
SpO2 (%)								
T0	97.73±2.69	97.59±1.69	0.015	0.902	8.832	<0.001	0.520	0.835
T1	98.81±1.27	98.48±1.72[a]						
T2	99.23±1.45	99.37±1.08[a]						
T3	99.31±1.12[a]	99.48±1.04[a]						
T4	99.58±0.92[a]	99.67±0.83[a,b]						
T5	99.46±1.14[a]	99.67±0.88[a,b]						
T6	99.50±1.07[a]	99.59±0.89[a,b]						
T7	99.46±0.99[a]	99.33±1.69[a]						
T8	98.77±1.70	98.41±1.95[f,g]						

Note:.
[a] , compared with T0, $P < 0.05$;.
[b] , compared with T1, $P < 0.05$;.
[c] , compared with group T2, $P < 0.05$;.
[d] , compared with T3, $P < 0.05$;.
[e] , compared with T4, $P < 0.05$;.
[f] , compared with T5, $P < 0.05$;.
[g] , compared with T6, $P < 0.05$;.
[h] , compared with C group, $P < 0.05$. T0: admission; T1: before anesthesia induction; T2: during intubation; T3: before skin incision; T4: skin incision; T5:30 min after skin incision; T6:60 min after skin incision; T7: completion of surgery; T8: upon leaving the recovery room.

has attracted significant research attention.

The advantages of TEAS

Transcutaneous Electrical Acupoint Stimulation (TEAS) is a non-invasive neuro-electroacupuncture therapy, combines traditional acupuncture theory with modern transcutaneous electrical nerve stimulation technology, demonstrating broad clinical application potential. TEAS stimulates specific acupoints with electrical current to generate neural modulation effects along meridians, aiming to reduce central and peripheral inflammatory responses, inhibit microglial cell activation, and prevent neuronal apoptosis, thereby exerting neuroprotective effects [18,19]. It is simple to operate, effective, and has minimal side effects. Personalized treatment can be achieved by

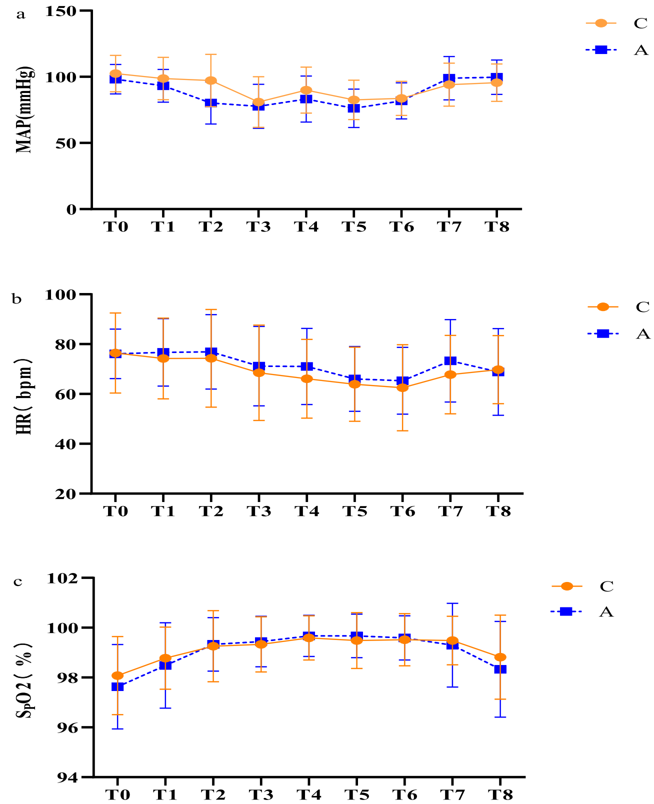


Fig. 6. Changes of MAP, HR, and SpO2 at different times during the operation in the two groups
Note: Fig.s a, b, and c show the changes of MAP, HR, and SpO2 of the two groups at different time points during operation, respectively.

Table 5
Comparison of intraoperative hemodynamic stability, postoperative recovery room residence time, and postoperative extubation time between the two groups of patients (n = 53).

Event	C	A	T-test/ X^2-test	
			t/X^2	P
Hemodynamic stability				
Hypotension (MAP<80 %)	13 (50.0 %)	9 (33.3 %)	5.681	0.735
Hypertension (MAP>120 %)	4 (15.4 %)	6 (22.2 %)	4.615	0.329
Bradycardia (HR<48 bpm)	3 (11.5 %)	3 (11.1 %)	4.325	0.375
Tachycardia (HR>90 bpm)	2 (7.7 %)	6 (22.2 %)	8.231	0.083
Extubation time (min)	10.85 ±8.44	7.44 ±6.03	1.693	0.097
Recovery room residence time>30min	11 (42.3 %)	8 (29.6 %)	0.926	0.336

Note: If hypotension, hypertension, bradycardia, or tachycardia persist for >5 min, they will be recorded as a single occurrence.

precisely controlling the timing of acupuncture and electrical stimulation parameters, such as duration, intensity, and frequency.

In perioperative management, TEAS has been widely applied to various surgical patients, showing significant effects not only in organ protection, inflammation alleviation, pain management, and stress reduction but also in actively improving PND [20]. Furthermore, TEAS has been successfully applied in the rehabilitation treatment of patients with cerebrovascular diseases such as cerebral infarction and Alzheimer's disease, achieving satisfactory therapeutic outcomes, and further validating its value in neuroprotection [5,21].

Selection of acupoints

Acupoints form the therapeutic cornerstone of TEAS for managing PND. The selection of acupoints follows to the "Xingnao Kaiqiao" principle in traditional Chinese medicine, with a typical protocol involving electrical stimulation at Baihui (GV20), Neiguan (PC6), Zusanli (ST36), and Sanyinjiao (SP6) bilaterally [18]. Baihui (GV20), located on the governor vessel meridian, is essential for treating neurological and psychological disorders, providing neuroprotective and sedative effects. Neiguan (PC6), a crucial acupoint, is highly effective for nervous system and mental health disorders and also plays a significant role in managing gastrointestinal diseases. Zusanli (ST36) is known for enhancing gastrointestinal motility and improving digestion, with potential benefits for brain function recovery. Sanyinjiao (SP6), located at the intersection of the liver, spleen, and kidney meridians, promotes blood circulation, mental tranquility, and digestive function. This study employs TEAS at these acupoints to evaluate its efficacy in preventing PND.

Analysis of main indicators

PND encompasses a spectrum of cognitive changes that can manifest both prior to and within the first 12 months following surgery. Clinically, PND is categorized into several distinct types: pre-existing cognitive impairment, preoperative delirium, postoperative delirium within the first week, delayed neurocognitive recovery, and post-operative PND as defined by specific criteria [22].

For accurate identification of PND, objective cognitive assessment is paramount, with neuropsychological testing playing a critical role in evaluation. MMSE is a widely recognized screening tool for PND. This assessment evaluates neurocognitive function across five key domains: orientation, attention, calculation, recall, and language.

A diagnosis of PND is typically established when there is a post-operative decline of two or more points in the MMSE score compared to the preoperative baseline [11]. In light of this, our study employed MMSE scores to meticulously assess the cognitive function of elderly patients in the perioperative period.

Previous research has consistently shown that perioperative neurocognitive decline is a significant issue following hip fracture surgery in elderly patients [23]. Our study's findings indicated no significant pre-operative difference in MMSE scores between the two groups ($P > 0.05$). However, both groups experienced a decline in MMSE scores on the 1st and 3rd postoperative days compared to preoperative levels, which was statistically significant ($P < 0.05$). By the 7th postoperative day, MMSE scores had returned to preoperative levels, with no significant differences observed. Notably, Group A had a significantly lower incidence of PND than Group C on both the 1st and 3rd postoperative days ($P < 0.05$). These results suggest that while neurocognitive function is temporarily affected in the immediate postoperative period for both groups, there is a gradual recovery towards preoperative levels by the 7th day, highlighting the impact of anesthesia and surgical stress on cognitive function. Intriguingly, TEAS has been linked to a reduced incidence of PND in elderly patients, suggesting its potential to aid in cognitive recovery.

There is substantial evidence that surgery and anesthesia can trigger a robust systemic inflammatory response and activate the immune system, leading to an inflammatory reaction within the central nervous system. These inflammatory mediators can damage brain cells by disrupting the blood-brain barrier and releasing harmful substances such as proteolytic enzymes, oxygen-free radicals, and lysosomal enzymes, resulting in neuronal damage and impaired synaptic plasticity [10]. In this study, a significant increase in IL-1β levels was observed 30 min post-surgery for both groups compared to preoperative levels ($P < 0.05$). Group A exhibited a more pronounced decrease in IL-1β levels on the first postoperative day. The interaction between time and group was statistically significant for IL-1β ($P < 0.05$). Additionally, group C exhibited a significant increase in IL-6 levels at 30 min and the 1st day post-surgery, whereas group A did not show a significant increase at

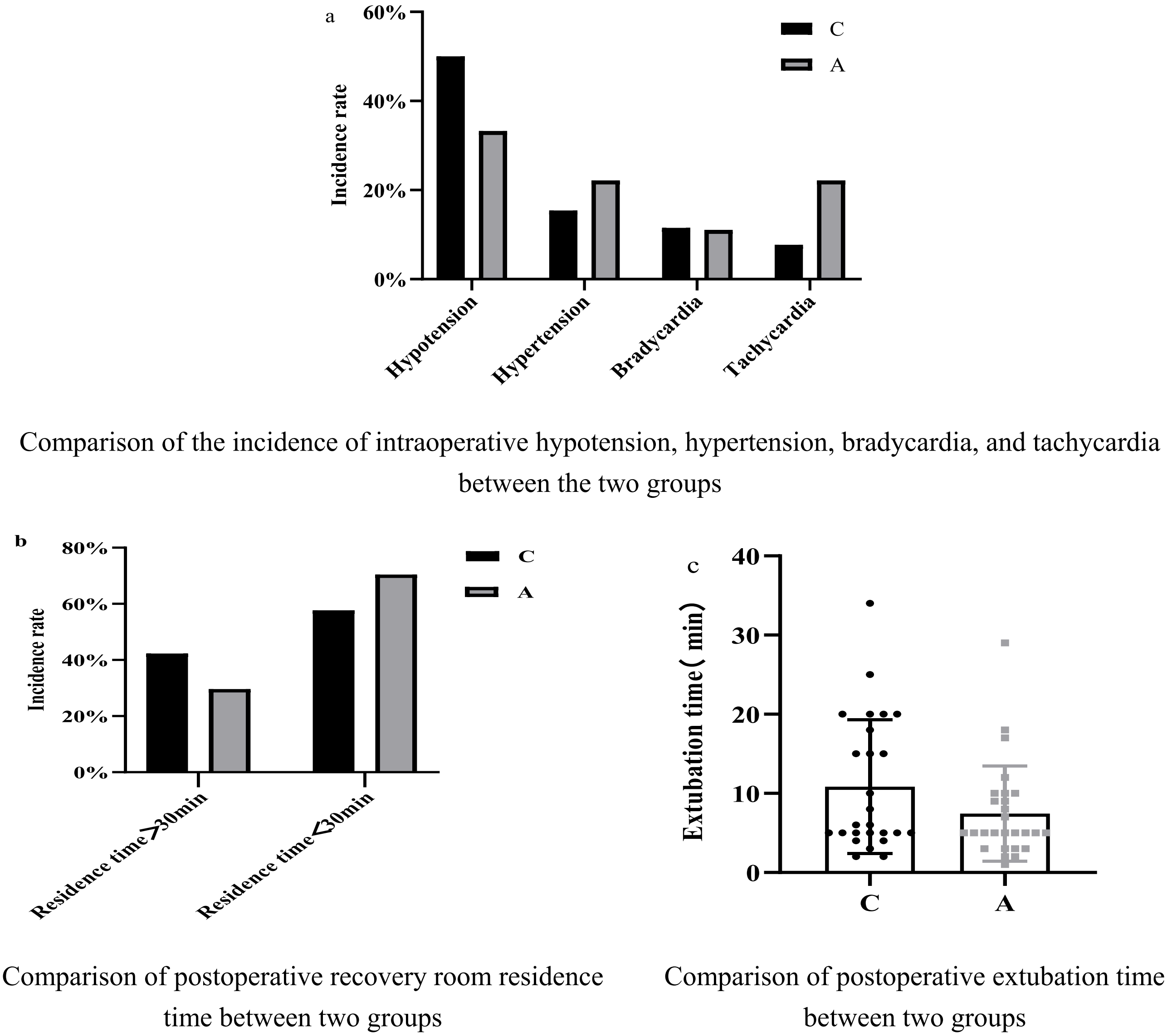


Fig. 7. Comparison of intraoperative hemodynamic stability, postoperative recovery room residence time, and postoperative extubation time between the two groups

Note: Panel a shows the incidence of intraoperative hypotension, hypertension, bradycardia, and tachycardia between the two groups. Panel b shows the postoperative recovery room stay duration between the two groups. Panel c shows the extubation time between the two groups.

Table 6

Comparison of postoperative adverse reactions between the two groups ($n = 53$, %).

Adverse reactions	C	A	X^2-test	
			X^2	P
Digestive system				
PONV	8 (30.8 %)	1 (3.7 %)	5.097	0.024
stomach pain and distension	2 (7.7 %)	3 (11.1 %)	<0.001	1.000
Cardiovascular system				
Hypotension	1 (3.8 %)	3 (11.1 %)	0.231	0.631
Hypertension	2 (7.7 %)	3 (11.1 %)	<0.001	1.000
Tachycardia/bradycardia	2 (15.4 %)	2 (11.8 %)	<0.001	1.000
Respiratory system				
Hypoxemia	4 (15.4 %)	0 (0.0 %)	2.559	0.110
Central nervous system				
headache and vertigo	5 (19.2 %)	5 (18.5 %)	<0.001	1.000
Delirious agitation	3 (11.5 %)	5 (18.5 %)	0.106	0.745
Chills and Fever	5 (19.2 %)	4 (14.8 %)	0.004	0.950

these time points. The interaction effect between time and group was again statistically significant in IL-6 ($P < 0.05$).

It is hypothesized that significant surgical trauma and a robust stress response may precipitate PND by activating the central nervous system's inflammatory response through inflammatory mediators. TEAS appears to mitigate the expression of inflammatory factors such as IL-1β and IL-6, potentially inhibiting the inflammatory response and enhancing neuronal function and neurocognitive performance. This aligns with previous findings that surgical trauma can trigger M1 polarization of microglia in aged mice, leading to hippocampal neuroinflammation and postoperative cognitive impairment [24],which is consistent with the observed increase in postoperative inflammatory factors in this study. Prior studies have demonstrated that TEAS can effectively reduce the expression of inflammatory factors such as IL-6 and C-reactive protein in elderly patients after surgery, thereby inhibiting the inflammatory response and reducing the incidence of PND [13]. These findings underscore the role of TEAS in modulating inflammatory factors and improving postoperative cognitive function, thereby corroborating the results of our study.

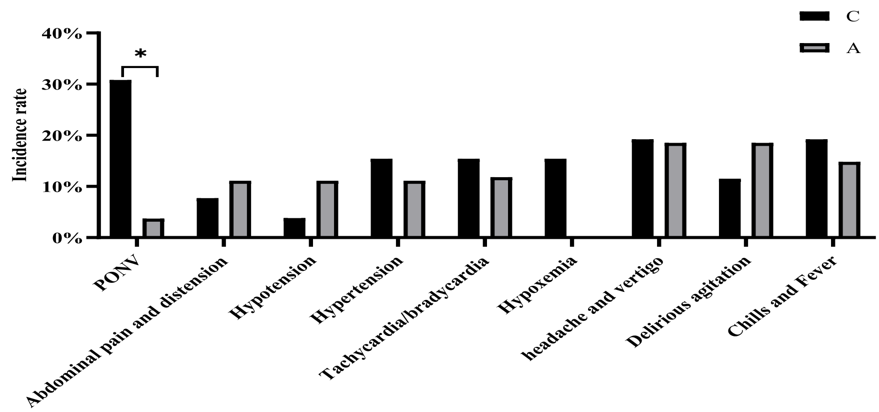


Fig. 8. Comparison of postoperative adverse reactions between the two groups
Note: Compared with group C, * $P < 0.05$.

Table 7
Comparison of postoperative VAS scores, PCIA compression times, drug consumption, and hospital stays between the two groups ($n = 53$, $\bar{x}\pm s$).

Event	C	A	T-test	
			t	P
VAS scores				
24 h after surgery	3.96±0.96	2.78±0.97	4.458	<0.001
48 h after surgery	2.65±0.94	1.85±0.72	2.876	0.001
PCIA compression times	8.62±1.84	7.11±2.14	2.745	0.008
drug consumption (ml)	89.23±3.67	86.22±4.27	2.745	0.008
hospital stays (days)	14.77±3.57	13.37±3.03	1.541	0.130

The S-100β protein, predominantly found in astrocytes and Schwann cells of the central nervous system, serves as a promising biomarker for neuronal injury [25]. In this study, a significant increase in S-100β protein levels was observed in both groups 30 min post-surgery compared to preoperative measurements, with levels gradually reverting to baseline by the first postoperative day. Notably, no significant inter-group differences were observed in S-100β protein levels.

Prior research has documented a transient elevation in S-100β levels in serum and cerebrospinal fluid within four hours post-surgery, normalizing by 24 h [26]. Interestingly, in this study, TEAS intervention did not diminish the expression of S-100β. This lack of effect might be due to factors such as variability in participant selection, surgical techniques, genetic predispositions of the subjects, and disparities in

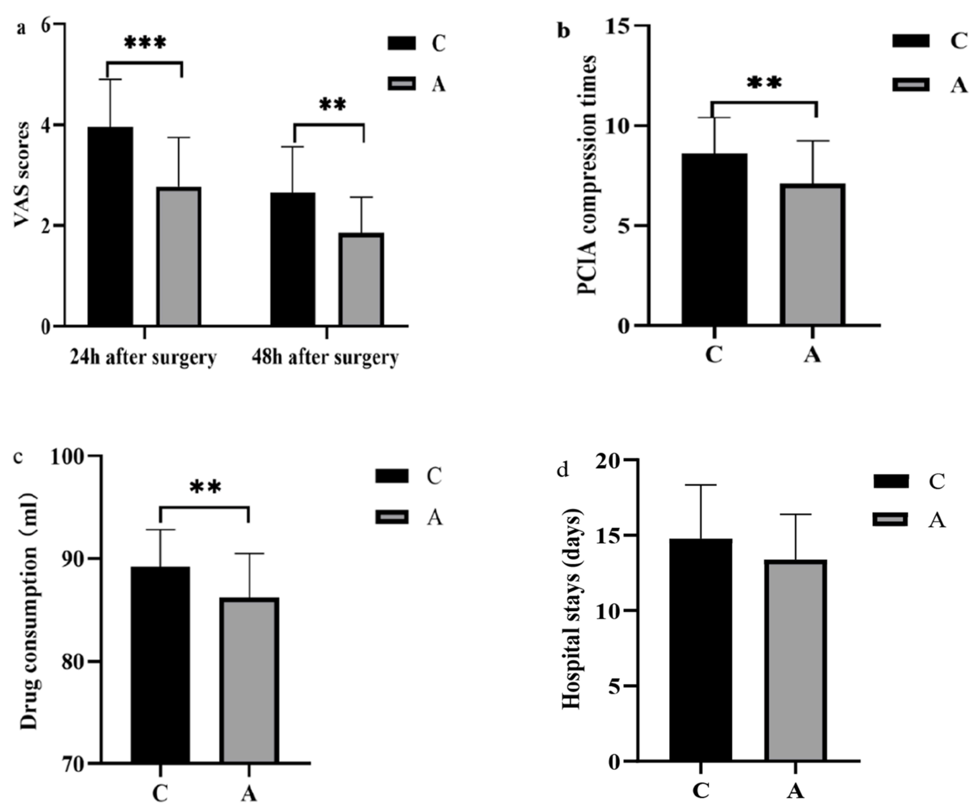


Fig. 9. Comparison of VAS scores at 24 and 48 h after surgery, PCIA compression times, drug consumption, and hospital stays between the two groups.
Note: Panel a is the comparison of VAS scores 24 and 48 h after surgery between the two groups; Panel b is the comparison of the total number of PCIA compressions 48 h after surgery between the two groups; Panel c is the comparison of the total PCIA drug consumption 48 h after surgery between the two groups; Panel d is the comparison of length of hospital stay between the two groups. Compared with group C, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

PND assessment methodologies.

Analysis of secondary indicators

The development of hypertension is intricately tied to the regulatory functions of the hypothalamus within the central nervous system. Hypertensive reactions are commonly triggered by external noxious stimuli, such as intubation and skin incision. Our observations of MAP indicate that TEAS application is associated with a diminished stress response. In this study, Group A experienced a less pronounced increase in MAP during these procedures compared to Group C. Extensive previous research has demonstrated that acupuncture therapy can lower blood pressure through mechanisms involving the renin-angiotensin system, vascular endothelial function, oxidative stress, and neuroendocrine regulation [27]. These findings suggest that TEAS can mitigate cardiovascular responses to surgery while maintaining patient safety.

Moreover, no significant differences were observed between the two groups in terms of intraoperative hypotension, hypertension, bradycardia, or tachycardia ($P > 0.05$), indicating that TEAS does not negatively impact hemodynamic stability during surgery. Our study also found a significant reduction in the incidence of PONV in Group A compared to Group C ($P < 0.05$). Prior research has shown that TEAS can hasten recovery after surgery and decrease the occurrence of PONV [28], which aligns with our results. The use of TEAS may reduce the likelihood of PONV by activating adrenergic receptors and regulating the release of endogenous 5-hydroxytryptamine [29]. Additionally, Group A had significantly lower VAS scores at 24 and 48 h after surgery than Group C. Furthermore, the required doses of PCIA within 48 h postoperatively were significantly reduced in Group A ($P < 0.05$). These results indicate that TEAS intervention effectively alleviates postoperative pain, corroborating numerous reports in the existing literature. Some studies have further confirmed the efficacy of TEAS in controlling postoperative pain following major spinal surgery, proposing its potential as an adjunct for short-term pain management [30]. The analgesic mechanism of acupuncture is thought to involve the inhibition of pain signal transmission, reduction in the production of endogenous algogenic substances, and the facilitation of endogenous opioid peptide release, thereby inducing analgesia [31].

Limitations

This study, while demonstrating the potential benefits of EA in reducing PND in elderly hip fracture patients, has several limitations. The small sample size limits generalizability and the detection of subtle differences. Future studies require a larger sample size to elaborate on and validate the effects of EA. Additionally, the one-week follow-up period is insufficient for assessing long-term cognitive outcomes. Prolonged follow-up is needed to fully evaluate EA's sustained effects on cognitive recovery and function. Further research is essential to address these limitations and elucidate the mechanisms and long-term benefits of EA intervention.

Conclusion

This study demonstrates that preoperative and postoperative electroacupuncture intervention is beneficial in reducing the incidence of PND in elderly patients with hip fractures. Furthermore, it exhibits distinct advantages in attenuating the release of inflammatory factors, stabilizing hemodynamics, alleviating postoperative pain, and mitigating PONV.

Clinical trial number and registry URL

The trial was registered before patient enrollment at chictr.org.cn (ChiCTR2100053050, principal investigator: Wei Chen, date of registration: November 10, 2021, https://www.chictr.org.cn/showproj.html?proj=136703).

Ethical statement

The study was approved by the appropriate Institutional Review Board (IRB), and written informed consent was obtained from all participants. This study was approved by the Ethics Committee of the Second Affiliated Hospital of Zhejiang Chinese Medicine University (IRB: 2019-KL-129–01).

CRediT authorship contribution statement

Jiaqi Long: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaoming Zhang:** Writing – review & editing, Conceptualization. **Jialing Zhang:** Methodology, Investigation, Data curation. **Tao Zhou:** Methodology, Investigation, Data curation. **Wei Chen:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Yuebing Li:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no conflicts of interest.

Funding

Zhejiang Traditional Chinese Medicine Scientific Research Fund Project [2020ZB102] and Jiangxi Provincial Health Commission Technology Plan [202510679].

Acknowledgements

We thank all the study participants who enthusiastically joined the study.

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